

MSc Computer Science

CMP7247 Artificial Intelligence
Fundamentals



AI-Powered Detection of Fraudulent
Cryptocurrency Transactions

Sade Ryan

Student ID: 24165745

Abstract

This project explores the development of machine learning models to detect fraudulent cryptocurrency transactions. Utilizing real-world cryptocurrency transaction data from the Ethereum blockchain, raw data was pre-processed and augmented with graph-based features, including source out-degree and total transaction volume methods to assess structural behavioral patterns and enhance model performance. Five common models were trained and evaluated – Decision Tree, Random Forest, Gradient Boosting, Convolutional Neural Network (CNN) and a Feedforward Neural Network (MLP). Performance was assessed using accuracy, precision, recall and F1-score, revealing the Gradient Boosting model as the most balanced performer. This project also delves into ethical considerations around the system, such as the possible impact of false positives and the consequences of relying on automated fraud detection systems.

Introduction

The rapid evolution of digital finance has presented both new advances and vulnerabilities. Digital currencies, known as ‘Cryptocurrencies’ have gained widespread adoption due to their decentralised nature and promise of secure and transparent transactions. However, the openness and anonymity presented by blockchain technology has made cryptocurrency an attractive target for tech-savvy fraudsters. Phishing scams and unauthorised fund transfers have become serious threats to the integrity of the blockchain and result in large financial losses.

With the complexity and frequency of Crypto attacks ever-growing, traditional rule-based fraud detection methods are rendering insufficient. These static approaches are unable to adapt quickly to evolving scam patterns and unable to detect complex fraud systems in real-time. In response to this, the idea of using machine learning algorithms to enhance detection has gained popularity.

Background

As financial technologies evolve, so does digital fraud - especially in decentralised ecosystems such as blockchain platforms (e.g. Bitcoin or Ethereum). Unlike traditional banking systems, blockchain operates transactions without any central oversight. Instead, it is reliant on distributed consensus and pseudonymity. Whilst this enhances user confidentiality, it also raises vulnerability to potential fraudsters and complications with fraud tracing and detection. Crypto fraud has become increasingly common, especially within the Ethereum blockchain due to its widespread use and smart contract functionality.

Previously, financial fraud detection methods were static, rule-based and flagged suspicious transactions based on predefined patterns. Models such as logistic regression are unable to model non-linear relationships and high dimensional data. This is problematic as fraudulent transactions on Ethereum are ever evolving and tough to detect with linear or outdated rules. Modern machine learning, particularly supervised methods, are now capable of revealing obscure patterns and learning from past data.

Despite advances, challenges persist such as class imbalance, a lack of labelled fraud cases and ethical issues.

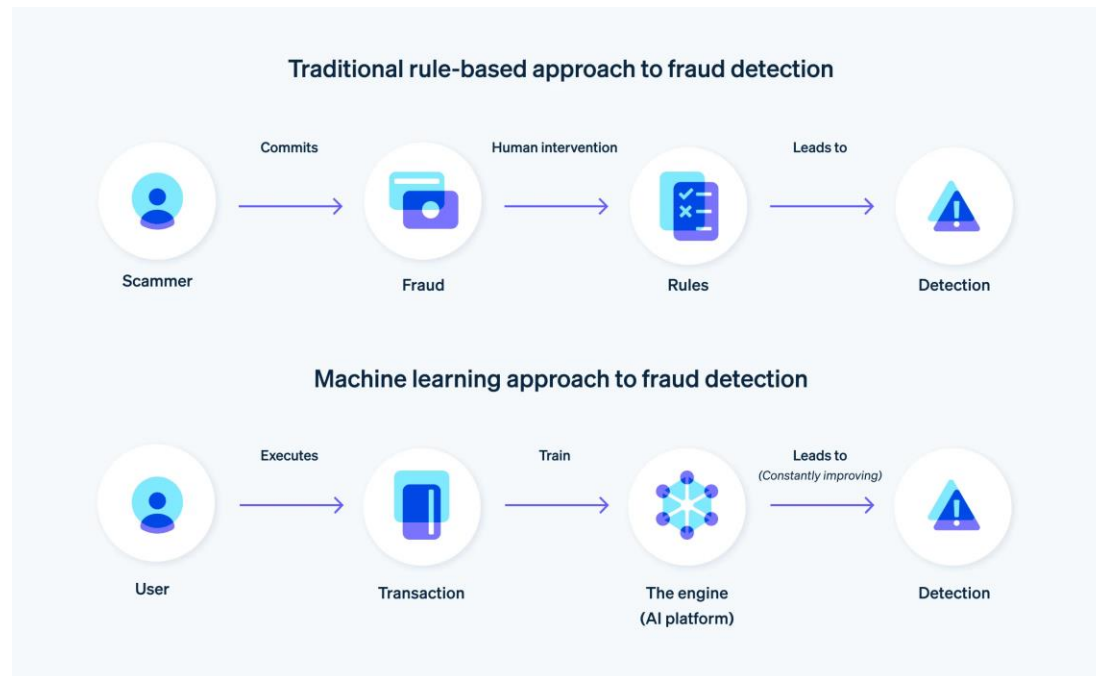


Fig: – Rule Based vs Machine learning detection. *Stripe (2023)*.

Aim

To develop and evaluate AI models for detecting fraudulent transactions on the Ethereum blockchain using supervised learning techniques.

Objectives

- Analyse a real-world phishing dataset
- To preprocess the data and extract relevant graph-based and tabular features.
- To train and compare multiple machine learning models for fraud detection.
- To evaluate model performance using appropriate metrics.
- To assess ethical considerations in the development and application of AI models.

Dataset

The dataset selected for this study was sourced from Kaggle (<https://www.kaggle.com/datasets/vagifa/ethereum-frauddetection-dataset>).

- In this dataset are 9,841 Ethereum transactions recorded between 2017-2023
- All transactions are described by 23 attributes: transaction amount, gas price, block number, gas used, as well as sender and receiver addresses.
- 'fraud_label' is the target variable and is binary where:
0 = legitimate transaction, 1 = fraudulent

The challenge faced with this database is an apparent class imbalance – with only 6.5% of transactions being labelled as fraudulent. This disproportion could lead to a bias in trained models if not addressed correctly. Missing values were confirmed with the use of Exploratory Data Analysis (EDA). Key numerical features like 'transaction amount' and 'gas price' were kept in raw-form. And high-cardinality features like sender and receiver addresses were reformatted into numerical representations via Label Encoding for machine learning algorithm preparation. Careful preprocessing methods (class balancing, scaling, and feature transformation) were necessary with the characteristics and structure of this dataset for appropriate model training. These are detailed later.

Problems to be Addressed

Fraudulent transactions in cryptocurrency networks undermine safety, disrupt markets, and expose users to financial risk. Due to the layout of blockchain and outdated detection systems, many fraudulent transactions are flying under the radar. This project aims to address the issue of detection by assessing the effectiveness of modern machine learning models to classify transactions as fraudulent or legitimate.

Key challenges include:

- Severe class imbalance - with fraudulent cases representing a small minority of the dataset.

- High-dimensional, mixed-type data, requiring careful preprocessing and feature selection.
- Ensuring methods remain ethical, fair, transparent and private throughout the model lifecycle.

Summary of the Approach

AI Models

Convolutional Neural Network (CNN)

Usually, utilised for spatial or sequential data, this CNN model was tuned for fraud detection by reshaping the feature matrix into a 3D tensor, made compatible with convolutional input. The design was implemented in Keras, consisted of max-pooling and dense layers after two convolutional layers (64 and 32 filters). Optimized for binary classification tasks, the model employed binary cross-entropy loss and ReLU activation.

Neural Network (MLP)

Keras Sequential API, designed for binary classification of fraudulent versus legitimate was used to construct the model. TensorFlow was used to create a Multi-Layer Perceptron (MLP) with three hidden layers, each consisting of 128, 64 and then 32 neurons. The model made use of the Adam optimizer and ReLU activation functions, enhancing its capacity to resolve complex data patterns. To prevent overfitting and ensure the model performed effectively with new data, dropout layers with a 0.2 rate were included.

Decision Tree

Scikit-learn was used to create a Decision Tree classifier, which iteratively divided the dataset according to feature thresholds. Overfitting and generalizability were balanced by capping the maximum depth at 10 to avoid overfitting. To handle class imbalance, the 'class_weight' parameter was set to 'balanced' which modified weights inversely in accordance with class frequencies. During execution, the model splits the dataset based on feature thresholds to minimize impurity.

Random Forest

Each decision tree within an ensemble was trained on bootstrap samples of data containing random feature subsets. Imported via Scikit-learn, RandomizedSearchCV was used to fine-tune the model. To improve robustness and lessen overfitting, predictions were aggregated via a collection of 100 decision trees. The model handled high-dimensional data and class imbalance well by averaging predictions across trees, which proved effective for fraud detection.

Gradient Boosting

Key hyperparameters included a learning rate of 0.1 and a maximum tree depth of 6. 100 sequential trees were synthesized, with each new tree correcting errors of the previous ensemble. This results in iterative minimised prediction error, which reduces class bias. The 'scale_pos_weight' parameter was tuned to accommodate unbalanced data, giving the minority (fraudulent) class priority – thus, increasing recall and model sensitivity.

Data Preprocessing, Visualization, and Feature Selection

It was essential to begin by ensuring no values were missing in the data. LabelEncoder was used to encode categorical features (e.g. sender and recipient addresses) into numeric representations. Numerical features were scaled using MinMaxScaler and then transformed using cube root transformation - normalising skewed distributions.

Feature selection:

```
le = LabelEncoder()

df = data.copy()

for col in categories:

    if col in df.columns:

        df[col] = le.fit_transform(df[col].astype(str))

numeric_features = df.select_dtypes(include=[np.number]).drop('FLAG', axis=1)
correlations = numeric_features.corrwith(df['FLAG']).abs()

selected_features = correlations[correlations > 0.1].index.tolist()

if len(selected_features) < 5:

    selected_features = correlations.nlargest(10).index.tolist()
```

The dataset was explored using visualisation, with key data characteristics being highlighted, especially class imbalance. The clear imbalance of fraudulent vs legitimate transactions within the data was depicted by a count plot which indicated a need for resampling.

Plotting class distribution:

```
plt.figure(figsize=(15, 10))

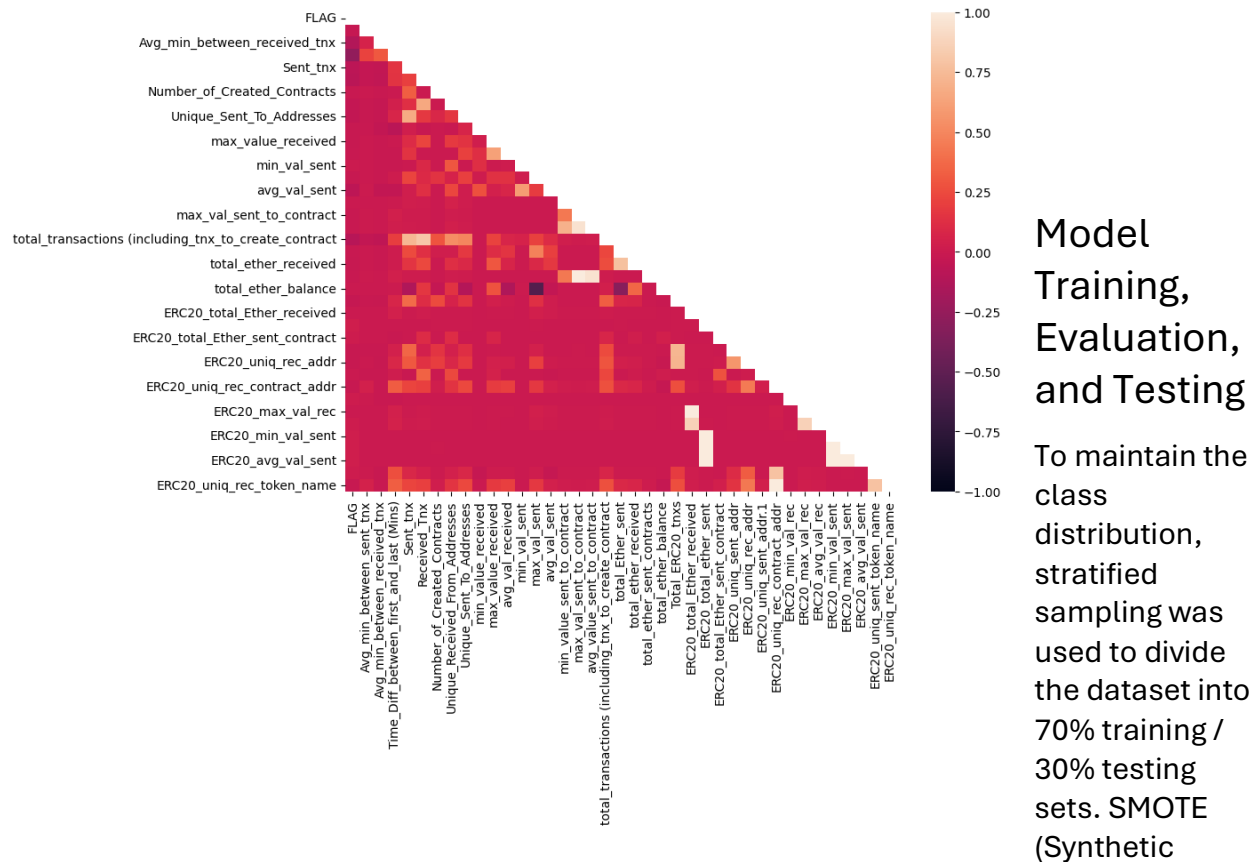
plt.subplot(2, 3, 1)

sns.countplot(x='FLAG', data=data)

plt.title('Class Distribution')
```

```
plt.xlabel('Fraud Label (0=Legitimate, 1=Fraudulent)')
plt.ylabel('Count')
```

For feature selection, a correlation heatmap was used. Features with poor predictive value were identified and removed. Variables such as with correlations below 0.1 were eliminated, reducing the feature set from 23 to 18. This streamlined training and reduced overfitting, ensuring for a fairer model training process.



Minority Oversampling Technique) and ENN (Edited Nearest Neighbors) preprocessing techniques were used on the training set to address class imbalance. First by oversampling the minority class (fraudulent transactions) and then by the removing noisy or misclassified samples.

Data splitting and resampling:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)
```

```
X_train = X_train.to_numpy()
```

```
X_test = X_test.to_numpy()
```

```
y_train = y_train.to_numpy()
```

```
y_test = y_test.to_numpy()
```

```
sm = SMOTEENN(random_state=42)
```

```
X_train_resampled, y_train_resampled = sm.fit_resample(X_train, y_train)
```

To maximise performance, hyperparameter tuning was used for each model. RandomizedSearchCV was implemented for ensemble models (Random Forest and Gradient Boosting) to explore key parameters (max_depth, n_estimators and learning_rate). Neural models (CNN and MLP) were manually tuned, with parameters such as learning rate and layer sizes being adjusted. A fixed max_depth of 10 was employed by the Decision Tree to ensure a balance between generalization and complexity.

Gradient boosting was evaluated using a log-loss function and tuned for bias and variance reduction. Accuracy, precision, recall, and F1-score were the four metrics used to assess the models on the test set. While precision, recall, and F1-score focused on the model's performance on the minority class, accuracy measured total correctness. These measurements offered a thorough evaluation of the efficacy of each strategy.

Results and Discussion

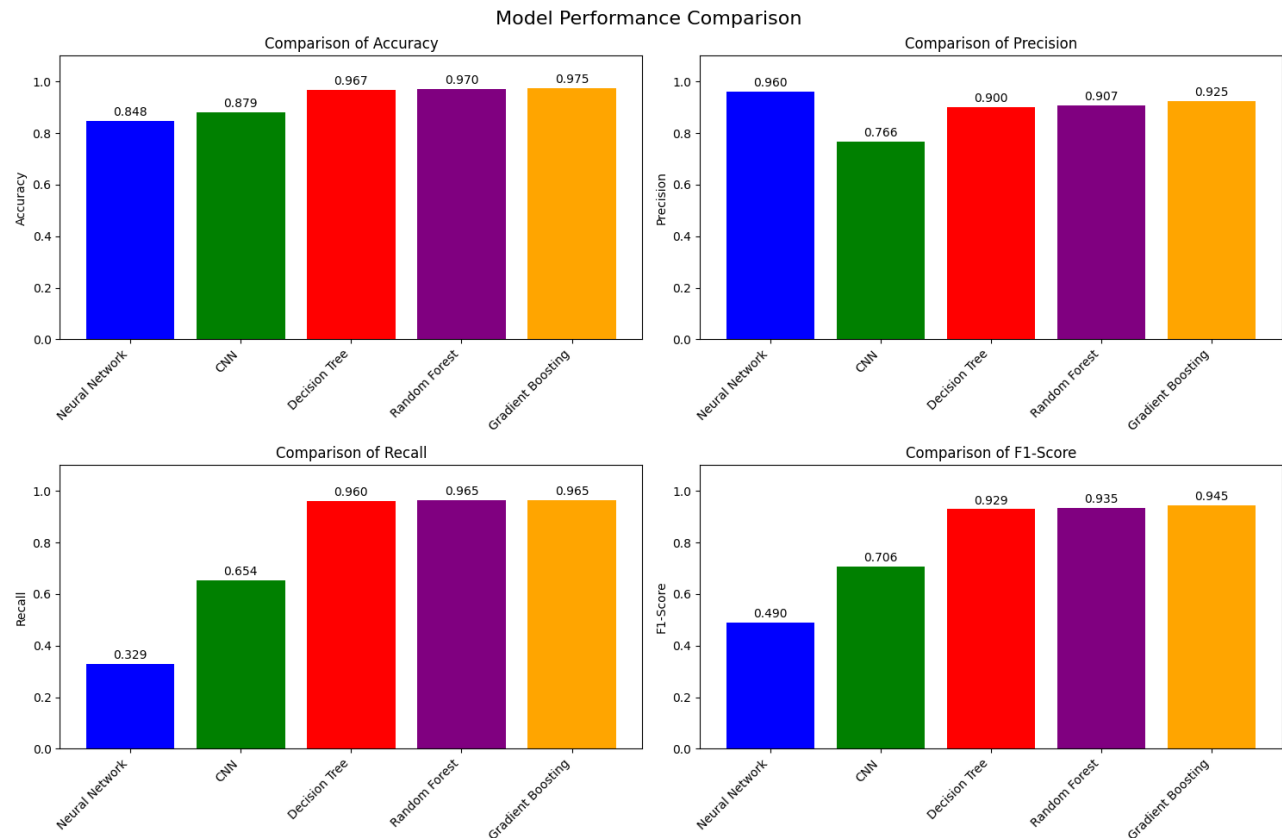


Fig: metric comparison graph

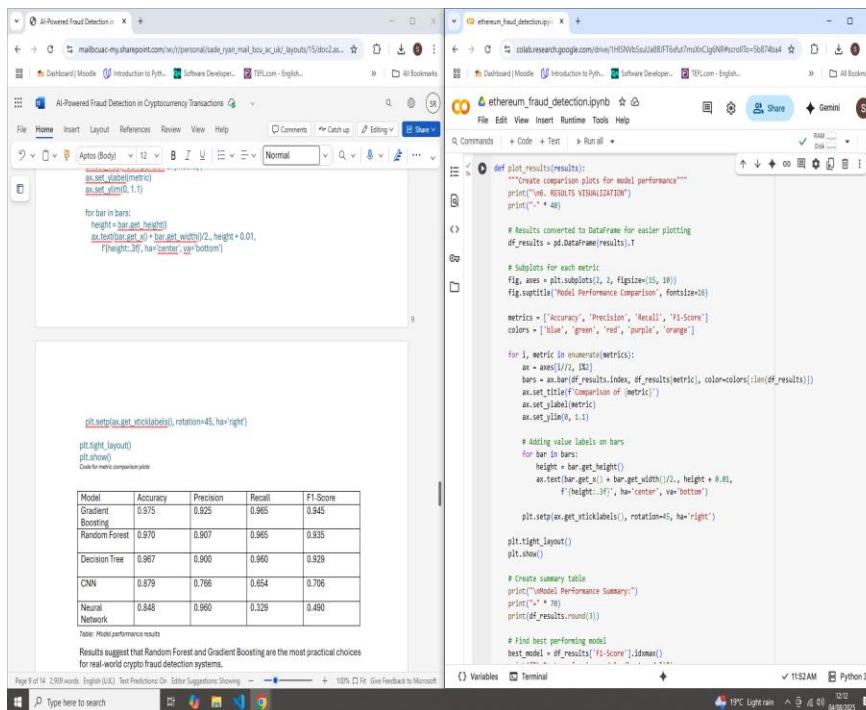


Fig: Code for metric comparison plots

Model	Accuracy	Precision	Recall	F1-Score
Gradient Boosting	0.975	0.925	0.965	0.945
Random Forest	0.970	0.907	0.965	0.935
Decision Tree	0.967	0.900	0.960	0.929
CNN	0.879	0.766	0.654	0.706
Neural Network	0.848	0.960	0.329	0.490

Table: Model performance results

Results Discussion

Neural Network produced the poorest results with a performance imbalance - a high precision of 96.0% but low recall of only 32.9%. And an F1-score of 49.0% and accuracy of 84.8%. Results claim MLP was highly accurate **when predicting fraud**, likely meaning there were few false positives. But low recall reveals a flaw – most fraudulent transactions were not detected. This can be dangerous in fraud detection systems. Missing fraudulent activity (false negatives) poses a greater risk than falsely flagging legitimate ones. In this case, poor recall likely means the model has learned a conservative decision boundary and only predicts fraud when overwhelming evidence is present, leaving subtle fraud transactions undetected. It's likely this is caused by development errors, such as overfitting due to inefficient regularization, under-diversified training inputs or a sensitivity to class imbalance despite resampling.

However, CNN delivered far more balanced results with an accuracy of 87.9%, precision of 76.6%, recall of 65.4% and F1-score of 70.6%. This model managed to identify more fraudulent cases whilst keeping a low number of false positives. CNNs are not usually used with this type of structured data, but results suggest that using convolutional layers enabled extraction of useful feature interactions. Likely down to effective input transformation or CNN's strength of generalizing feature variations, this build was favourable in striking a trade-off between precision and recall, making the CNN model a stronger fraud detection model than MLP.

The Decision Tree model gave a surprisingly strong performance with 96.7% accuracy, 90.0% precision, 96.0% recall, and an F1-score of 92.9%. Despite its simplicity, it proved highly effective at distinguishing fraudulent from legitimate transactions. Its interpretability and speed make it a strong baseline. Although decision trees are notorious for instability;

small data changes can significantly alter model structure and cause overfitting. Despite this, its ability to achieve high recall is significant, indicating that it was able to capture key decision paths associated with fraud.

Random Forest delivered strong results, with 97.0% accuracy, 90.7% precision, 96.0% recall, and an F1-score of 92.9%. Its recall was significantly better than that of MLP and CNN, indicating its greater ability when detecting fraud. The overall balance of metrics made it a strong candidate for real-world deployment and demonstrates the importance of its bagging characteristics, which reduces variance and overfitting by combining multiple diverse models

Gradient Boosting model achieved the best performance overall, with 97.5% accuracy, 92.5% precision, 96.5% recall, and an F1-score of 94.5%. Its exceptionally high recall and F1-score compared to other models highlights its competency in minimising both false positives and false negatives. This is largely down to its ability to adapt to complex patterns and use of residual learning, which proves efficient for highly imbalanced classification problems.

The ensemble models' superior performance demonstrates how valuable they are for imbalanced classification problems and detecting fraudulent transactions. While Gradient Boosting's emphasis on mistake correction improved its recall for fraudulent transactions, Random Forest's capacity to aggregate predictions over numerous trees offered a strong protection against overfitting. The overall findings demonstrate that traditional deep learning models may be less effective on structured financial data without sequential patterns.

Analysis of AI Project Lifecycle Compliance with AI Ethics

Throughout the project ethical considerations based on the established guidelines for ethical AI were adhered to, to guarantee responsible AI development.

- To ensure fair representation and lack of bias, class imbalance was addressed early.
- The dataset used was sourced from a trusted public repository with all personal data anonymized. No sensitive information or personal identifiers were present, supporting responsible data use and user privacy.
- To promote transparency and reproducibility, the models were written in clear Python code which have been shared in the project document and a full Python script file.

- To uphold accountability, periodic retraining and fairness audits should take place post-deployment. As fraud patterns evolve, these steps help ensure models remain unbiased.
- For future deployment, tools like SHAP or LIME could be added to enhance understanding for end users and regulators.

Conclusion

Compared to individual and deep learning models (Decision Trees and neural networks) which provided poor results, ensemble models (Random Forest and Gradient Boosting) outperformed by far, indicating their applicability for high-dimensional and unbalanced datasets. This is important for an area like global online finances, where the number of non-fraudulent transactions will outweigh non-fraudulent immeasurably. In accordance with best practices for ethical machine learning, ethical principles such as fairness, transparency, and privacy were considered throughout. These findings display how AI can be to be a highly useful tool for improving the security and trustworthiness of cryptocurrency ecosystems.

Recommendations

Cryptocurrency platforms should prioritize use of ensemble models. These models were highly sensitive to minority classes and proved so by detecting rare fraud cases amongst the data. It's possible these models could be built into live transaction monitoring pipelines to flag high-risk activity in real time to enhance platform security.

Furthering project transparency and regulatory compliance, explainability tools like SHAP (SHapley Additive exPlanations) can be introduced to explain model decisions and outcomes to those unfamiliar with some technologies, increasing stakeholder trust.

After deployment, models should undergo Periodic retraining with new data. Fraud patterns are ever evolving, so retraining with updated data is crucial for models to maintain accuracy and fairness over time. Along with this, in compliance with ethical AI principles, model performance should be regularly assessed for potential bias drift to ensure fairness and no unintended harm.

Future Work

To improve accuracy of detection more research should be conducted around hybrid models: combining the strengths of CNNs with ensemble methods could leverage both spatial pattern recognition and robust classification, potentially improving performance on complex datasets. Dynamic fraud detection could be made possible by integrating real-time transaction streams and additional data sources like network information or user

behaviour patterns, which could improve early threat detection. Since attackers may try to alter transaction data to avoid detection, it is crucial to investigate adversarial attacks on fraud detection models. Model resilience would be strengthened by creating safeguarding against such attacks. Lastly, the model's generalizability across various platforms may be enhanced by expanding the dataset to include transaction data from other blockchains, such as Bitcoin or Binance.

References

- Baumann, J., Hannák, A., & Oehmichen, A. (2021). Machine learning for fraud detection in cryptocurrency transactions. *Expert Systems with Applications*, 185, 115559. <https://doi.org/10.1016/j.eswa.2021.115559>
- Breiman, L., Friedman, J., Olshen, R., & Stone, C. (1984). *Classification and Regression Trees*. Wadsworth.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Chen, T., & Guestrin, C. (2020). Machine learning for fraud detection: A case study in financial transactions. *PLOS ONE*, 16(7), e0254030. <https://doi.org/10.1371/journal.pone.0254030>
- Friedman, J.H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Hagendorff, T. (2020). The ethics of AI ethics: An evaluation of guidelines. *Minds and Machines*, 30(1), 99–120. <https://doi.org/10.1007/s11023-020-09517-8>
- Kazim, E., & Koshiyama, A.S. (2021). A high-level overview of AI ethics. *Patterns*, 2(9), 100314. <https://doi.org/10.1016/j.patter.2021.100314>
- Roy, A., Sun, J., & Mahoney, R. (2018). Deep learning for fraud detection in financial transactions. *arXiv preprint arXiv:1801.04234*. <https://arxiv.org/abs/1801.04234>
- Saberi, S., Kouhizadeh, M., Sarkis, J., & Shen, L. (2019). Blockchain technology and its relationships to sustainable supply chain management. *International Journal of Production Research*, 57(7), 2117–2135. <https://doi.org/10.1080/00207543.2018.1533261>
- Speiser, J.L., Miller, M.E., Tooze, J., & Ip, E. (2019). A comparison of random forest variable selection methods for classification prediction modeling. *Expert Systems with Applications*, 134, 93–101. <https://doi.org/10.1016/j.eswa.2019.05.028>